IMPRINT

A SYMPHONY OF SPECIES

VOLUME 3, 2018- DEPARTMENT OF ZOOLOGY

The Xavier's Zoology Association

EDITORIAL TEAM

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EDITOR'S NOTE

Dear Readers,

What is communication? Communication can be thought of as the transfer of information from one organism to another. This transfer of information is ever present; since the creation of life, when two unicellular organisms accidentally bumped into each other to now, when we Skype people for a chat. With evolution, communication has not only advanced and increased in complexity, but has also diversified. The very second two signals indicate the same thing, one of the signals becomes moot. Hence, communication is an extremely unique process; it varies not only from kingdom to kingdom, or from species to species but also within a particular species.
And so, this year, with our theme - "A Symphony of Species", we would like to take you on a journey to explore the fascinating world of communication among organisms. Hopefully, we shall not only discover some novel interactions, but also deepen our understanding of the ones we commonly see around us.

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ASSOCIATION ACTIVITIES

WALK OF LIFE

The Walk Of Life returned in its second edition with an exhibition themed around Ecological niches. The rare specimens of the Zoology department were displayed according to the ecological niches they occupied alongside explanations and facts given by the members of the association.The exhibition was attended by several students and faculty from the different departments of the college as well as some students from other schools.

NAYAN KHANOLKAR

XZA started 2018 with an inspiring talk by Nayan Khanolkar, recipient of the BBC NHM Wildlife Photographer of the Year, 2016- Urban Wildlife Category. The audience listened with fascination as Nayan highlighted his journey from leisurely bird photography to gruelling, unpredictable and expensive Big Cat conservation. Mr. Khanolkar shared with us, the struggles and triumphs of shooting Mumbai's elusive and endearing big cat, the Indian Leopard, including setting up his own camera traps, working holistically with villagers and touring the world with his photo exhibitions.

JAMBULMAL TREK

The Xaviers Zoology Association went charging into the new semester with a long and challenging hike up the slippery slopes of the SGNP. We trekked through dense evergreen vegetation and waded through fresh water streams, all signs of the city forgotten, except for the calls of the familiar city crows. A myriad of crabs, spiders and dragonflies were spotted as the group made its way to the top, learning new things along the way.

PET TALK 101

Shivani Tandel, a renowned exotic pet veterinarian, presented a lecture about the nuances and intricacies of pet care. She not only highlighted the common ailments among our pets but also drew our attention towards the mental health of these animals. She also discussed career paths available for budding animal behaviourists and potential veterinarians.

KNOW YOUR FISH

Ketki Jog, a marine biologist working with "Know Your Fish"- a voluntary initiative towards an ocean friendly lifestyle, provided great insights to fish eaters with information regarding the statistics and periodicity of fishing, fish breeding and other valuable information for sustainable seafood consumption.

CHRISTMAS AT WSD

This year, the association decided to spend Christmas with man's most loyal friend. The welfare of stray dogs (WSD) an NGO, played host for this human-canine bonding. The visit started off with the WSD 'hoomans' briefing us on the history of the trust, their aim towards forming a healthy and happy street dog population, and the rescue and treatment of wounded stray dogs. This was followed by a tour through the center and fun activities like bathing, walking and feeding the dogs.

TETRAPACK RECYCLING

The association tied up with 'R U Reusing, Reducing and Recycling (RUR)' an NGO to recycle tetra packs into desks, books and bins. The association began this project with the goal towards making St. Xavier's a zero waste campus which promotes a sustainable living for all.

A SYMPHONY OF SPECIES

As the last rays of the sun disappeared over the canopy, a nightingale perched on the highest branch and began his song, the crickets below chirped merrily, a firefly began his beautiful dance of lights and a tigress chuffed goodnight to her little cubs. Nearby, a viper hissed as he watched an army of frogs croaking in the pond and a hyena laughed as she watched this little spectacle. An owl hooted, taking off into the night sky and slowly, the lights went out from the nearby villages as the humans tucked into their beds, singing their bedtime prayers.
Thus, began the Symphony of Species.

CONTENT

CALL OF THE CAROLINA CHICKADEE

Language – the most complicated communication system as a starting point


Carolina Chickadee

Human language is by far the most complicated communication system known. Despite the small number of distinct sounds in our languages and the fairly rigid rules that govern how those sounds are put together to make words and phrases, we are able to build utterances that can convey an infinite number of unique messages. In other words, languages are open-ended – they can convey a staggering and limitless amount of information.

Are there any non-human animal communication systems that share these key elements that make human languages so complicated? Are there any non-human animal signaling systems that comprise discrete units (sounds, visual displays, chemical components, etc.) with rules for how those units are put together to make longer signals, resulting in the possibility of open-ended communication?

We might first thinkof our closest living relatives – the great apes. There is evidence that great ape species such as chimpanzees possess large systems of vocal signals, but it is not yet clear that they structure these signals together in more complicated utterances. Outside of the great apes, other non-human primate species (for example, baboons or macaques or lemurs) tend also to have large systems of vocal signals, but only scattered evidence that those signals are composed in ways to produce longer and more complicated utterances.

Unexpected candidates for communicative complexity

Consider instead a group of species from an entirely different Class of animals – small songbirds of the family Paridae (the chickadees, tits, and titmice). My students and I have carried out a large number of studies on one of these species common in the southeastern United States – the Carolina chickadee (*Poecile carolinensis*). One thing true of chickadees and of many Paridae species in general is that they are highly gregarious and quite noisy, with large repertoires of vocal signals. Another thing true of chickadees and many Paridae species is that, where they occur, they are quite common (and so, practically, quite easy to study). A final thing to consider is that chickadees and many other Paridae species use a call that is one of the most structurally complicated vocal systems known outside of human language. In North American Paridae species, this vocalization is known as the chick-a-dee call, and is used throughout the year by both females and males in a wide range of contexts.

The chick-a-dee call of Carolina chickadees is made up of a small number of fairly distinct sound types, called notes. These notes vary in acoustic structure from relatively simple, tonal, and whistled elements to relatively complex, harmonic, and somewhat "noisy" sounds. Interestingly, these notes are put together according to fairly rigid rules of note ordering. Some notes are highly likely to be repeated if they occur in a call, whereas other notes almost always occur just once if they do occur. If a note type does occur in a call, furthermore, it tends to follow pretty rigid rules of note type ordering. (One interesting branch of the Paridae that warrants further study due to its apparently flexible and less rigid rules of note ordering in its calls is the black-lored tit group from India, *Machlolophus aplonotus* in the south and *Machlolophus xanthogenys* in the north). The result of these note type features of chick-a-dee calls is something that seems to share some core features of human languages – an open-ended signaling system.

It is worth pausing for a moment to consider this question of open-ended signaling systems in light of a far more heavily studied signaling system in songbird species. The songs used by songbirds (largely in the context of courtship and mating during the breeding season) can be relatively simple or highly complicated in acoustic structure, depending upon the species. However, for nearly all the species that have been studied, the repertoire of songs used by an individual is finite. Even species with ridiculously large song repertoires have a limited number of songs. If a researcher records an individual's songs for a long enough time period, eventually she or he will capture the entire song repertoire for that individual. As best we can tell, this is not the case with the chick-a-dee call – continued recording of the same individual will continue to reveal calls with novel note type composition, a crucial characteristic of an open-ended system.

So what are they chatting about?

As we have seen, the structure of the chick-a-dee call is quite complicated. There is enormous variation in note composition of these calls in chickadees and other Paridae species. As it turns out, this variation in note composition is not random. These birds do not appear to be generating complicated variation in their calls just for the sake of being complex (as might be the case with some songs of songbirds). Variation in note composition of chick-a-dee calls serves different functions. These birds vary their calls in biologically meaningful ways.

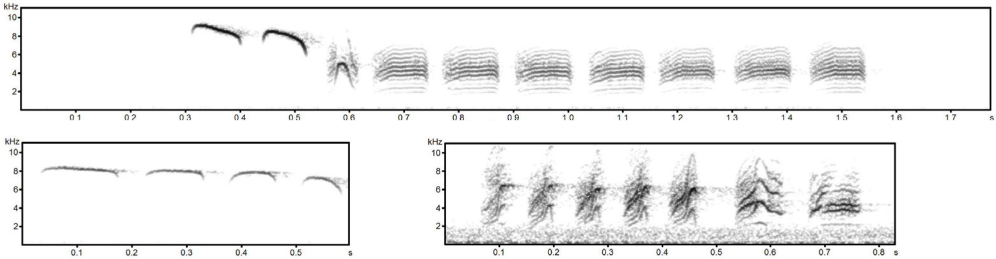
Perhaps the most important thing for a small songbird like a chickadee is to avoid becoming a meal of a predator. Chickadees, like other small prey species, have a suite of anti-predator behavior patterns, and the chick-a-dee call is one of these patterns. When individual chickadees detect a stationary or slow-moving predator in their environment, they tend to produce chick-a-dee calls that contain a large number of D notes (the dee-dee-dee notes that typically end these calls in many species). A large number of D notes in these calls seems to relate to the risk associated with a detected predator – the greater the risk (for example, from a smaller and more agile and lethal predator), the more Dnotes are used.

Producing a lot of D notes in chick-a-dee calls may help attract other species to the location of the signaler, which can be an effective mobbing strategy to drive the detected predator out of the area.

When a signaler produces a lot of A notes in its chick-a-dee calls, it likely communicates extreme risk to other flock members, but without drawing attention to itself. This is because high-pitched and pure tone vocalizations are quite difficult to localize in space.

Even more dangerous are situations where an individual detects an actively hunting predator. In these situations (for example, when a small hawk is hunting and flying rapidly near a flock of chickadees) the birds rarely produce D notes, but instead produce a large number of higher-pitched A notes. These introductory notes tend to have little frequency modulation – they sound like pure tone whistles at a very high frequency. When a signaler produces a lot of A notes in its chick-a-dee calls, it likely communicates extreme risk to other flock members, but without drawing attention to itself. This is because high-pitched and pure tone vocalizations are quite difficult to localize in space.

How about a much safer, but no less biologically relevant context? Chickadees and other Paridae spend most of their lives in social groups (breeding pairs and offspring in the summer and larger social flocks in the overwintering months). Birds are often out of visual contact with flock members as they move through their territories. How do flock members communicate about individual



Three sound spectrograms of chick-a-dee calls of Carolina chickadees. In each spectrogram, frequency (kHz) is on the Y-axis and time (seconds) is on the X-axis. The top spectrogram is representative of a call produced in a mobbing context. The left spectrogram is representaitve of a call produced in a high alarm / risk context. The right spectrogram is representative of a call produced in a flight context

movement given the importance of flock cohesion? We have found that at least for Carolina chickadees, individuals tend to produce a larger number of a third note type – the C note – in their calls when they are in the context of flight. This is an acoustically complex note type that would seem to be adaptive for providing information about signaler location, in that the notes have a rapid onset and wide frequency range.

Why are these calls so complicated?

Chickadees and many of their Paridae relatives have structurally complicated calls that also seem to be functionally complex. Why are these calls so complex? We do not yet have a definitive answer to this question, but one possibility has to do with the complicated social lives of these birds. Unlike most other songbird species, many species in the Paridae form relatively stable flocks in the overwintering months. This kind of social structure is fairly complex. Perhaps it is something about the complexity of their social lives that is driving the need for a complex communicative system? An experiment that manipulated flock size in Carolina chickadees found that birds placed into larger flocks used calls with greater diversity of note type combinations than birds placed into smaller flocks. More definitive studies to test this "social complexity hypothesis" are underway and hopefully we will have a more solid answer soon!

This "social complexity hypothesis" brings us back full circle to where we began – considering human language. Some researchers consider the question of language origins to be among the greatest unanswered questions in all of science. How did the first language come into being from a simpler communicative system? What was the adaptive need for such a complicated communicative system as language? Perhaps a greater understanding of what factors are driving complexity of chick-a-dee calls might help illuminate how language originated in our distant hominid ancestors.

Todd M. Freeberg is a Professor and Associate Head of the Department of Psychology at the University of Tennessee, U.S.A. He is an adjunct faculty member in the Department of Ecology and Evolutionary Biology at the university. He has been studying animal behavior and animal communication since his undergraduate days, and has been intrigued by chickadee vocal behavior since the late 1990s.

LET'S TALK ABOUT SEX

AND WHY DARWIN HATED PEACOCKS

If this title got you intrigued, you have proved an integral point I wish to make in this article. Animals want to communicate about sex, and they do so in the most vivid fashion. But apart from their showy exterior, they aim to communicate very subtle information about their fitness and that had Darwin stumped for the longest time.

Darwin once said that sight of peacocks made him sick, because while his theory of natural selection explained why Polar Bears have thick pelts of fur, the vivid plumage of a peacock's tail stood in open defiance to his theory of 'survival of the fittest'.

Darwin's theory of natural selection suggests that traits which are beneficial to an individual's survival would be the ones that get passed on through generations. In other words, the fittest individuals have the highest likelihood to reproduce. Why then, would something as cumbersome as peacock tail feathers ever evolve?

Peacock tail plumage is heavy. It consumes a lot of energy to grow, makes flying, walking and running a task and is hence, an impediment to the survival of the individual. Additionally, peahens continue to survive just fine without them.

Yet, natural selection continues to favour the existence of such a trait. This is not unique to peacocks and can be seen throughout the animal kingdom; from brightly coloured beetles and butterflies, to the elaborate songs and courtship rituals of birds and the conspicuous antlers on several species of deer. All possess traits which would baffle Darwin.

The fact that animals use ornamentation as a mechanism to communicate and signal their fitness to conspecifics has been recognised by several of his contemporaries but not Darwin himself.
Several models have been proposed to understand this such as Zahavi's Handicap Hypothesis,

Indian Peafowl

or the hypothesis put forward by Fischer proposing a runaway process. While they have some conflicting views on this topic, they are unified by the idea that these ornaments and elaborate mating rituals are a way for animals to communicate their relative fitness and abilities.

While exploring the Symphony of Species, we must pay attention to the subtle communication between animals, which is of immense evolutionary significance. Let us take a look at some examples to see how mating rituals and ornaments are integral to the process of natural selection

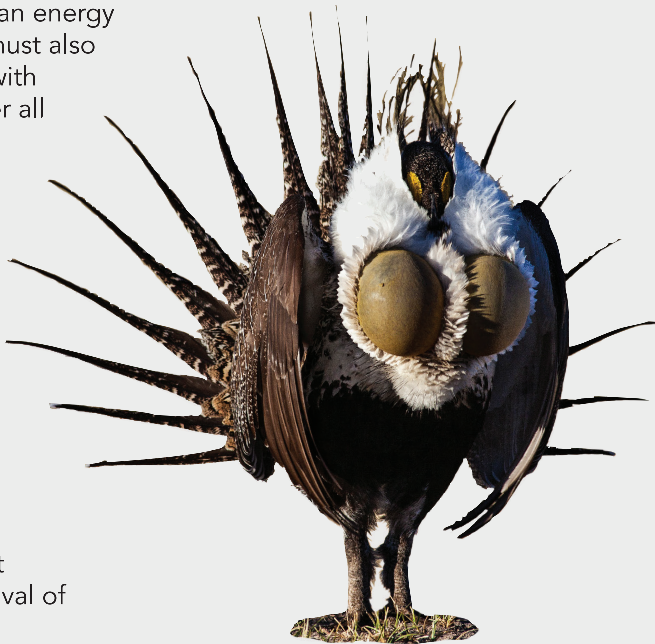
The Sage-grouse:

The sage grouse inhabits the vast, seemingly deserted plains of North America, where female counterparts are scattered far and wide. This means that attracting a potential mate is quite a task.

The males are endowed with a glamorous ruff of chest plumage under which, hide their coloured air sacs. The males congregate year after year at a 'Lek', where they begin their unique display. He inflates a pair of yellow air sacs, which protrude through the breast feathers during the display. Throwing his head back rapidly while deflating the air sacs, he lets out a low-pitched "plopping" sound. Females choose mates based on several criteria, including the quality of the "plopping."

Males have been reported to lose over a quarter of their body weight over course of this display (which may last around a month). It is clear that this is an energy intensive process. The males must also have enough energy to mate with almost 20 females per day after all their exertion.

While this dance may leave several males depleted of their energy resources, making them vulnerable to predators and leaving them in a state of weakness, it serves as a means for females to assess the stamina of their potential partners. This ensures that only the fittest and most durable males will be able to pass on their genes to the next generation, thus ensuring survival of the fittest.



Greater sage-grouse

The Widowbird:

Jackson's Widowbird

Another competition of stamina is seen in male Widowbirds that get new, impressive plumage and long tail feathers at the onset of mating season. They then make a 'dancing ring' amongst the tall grass and jump as high as they can. A male has to compete with many others doing the same in an area, while the spectators of the fairer sex gather to watch the games. The females judge the males not only how high they jump but also for how long. One by one, his competitors bow out of the competition.

This ritual serves as a way for females to test the fitness and endurance of the male, in turn allowing those favourable genes to be passed on to her offspring. On choosing a mate, she builds her nest in his dancing ring.

It was also seen by an experiment by Malte Andersson that the exaggerated tail length enabled males to be more visible and had evolved due to sexual preference for it (somewhat along the lines of the peacock).

The Flamingo:

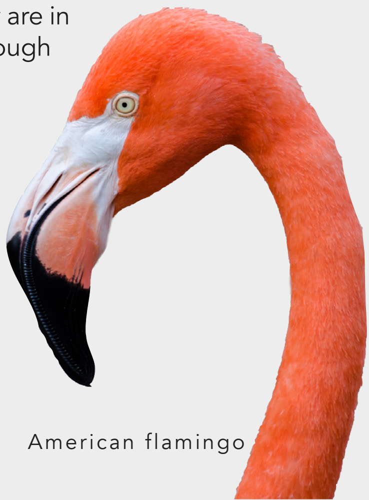
Nearly as well choreographed as Bollywood dance routines, flamingo males gather in a group and with swift left to right movements of their heads and intricate stomping movements of their legs, manage to grab the female's attention.

It has also been observed that the more colourful the males are, the higher the chance of them winning a mate. The vibrancy of the male gives the female a clue as to how adept at foraging he is, as the flamingo's colouration is a result of its diet which consists of algae and shrimp. Females thus get a chance to pick the best foragers for their mates ensuring efficient parental care for their offspring.

As one can see, these mating behaviours and physical ornaments seem to be a hindrance to an individual's survival. They are in accordance with the principle of natural selection through sexual selection.

Thus these seemingly ridiculous traits will continue to serve the purpose of communicating an individual's relative fitness. We can be sure that when it comes to sex, animals communicate their intentions with a proclivity for unbridled romance.

Zachary Borthwick, T.Y.B.Sc.



American flamingo

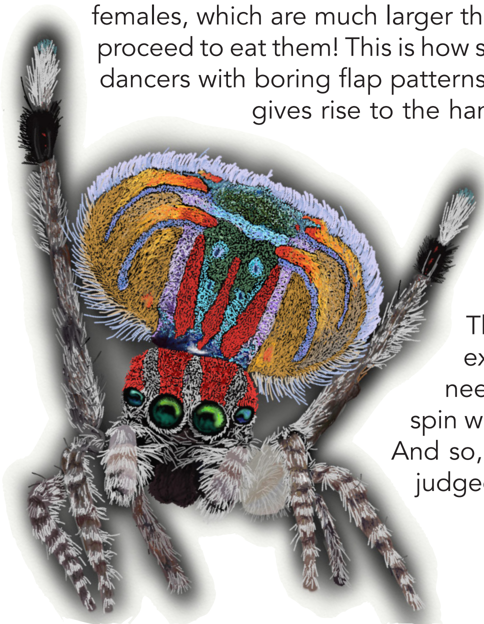
AN ORNATE DANCER

Mating and mating rituals form an important part of an organism's life and over time, the animal kingdom has taken upon itself, the task of developing a brilliant array of these. One of the most remarkable, complex and enchanting of mating rituals, belongs to a tiny critter, only a few millimeters long, even smaller than your fingernail!

The Maratus species, commonly known as the 'Peacock spider' is found only in some parts of Australia and China and belongs to the family of jumping spiders in the genus Maratus, which consists of around 44 species. Maratus males have evolved to carry out a complex series of visual and vibrational displays, all just to pique the interest of the fairer sex.

They start with vibrations which the female picks up from the ground (spiders don't have ears like we do). Once he has her attention, the visual spectacle begins. It involves a series of dance movements in which he raises his third pair of legs and also unfurls a flap present right over his opisthosoma or 'butt', if you may. This flap, which is similar to a fan is then waved in a back and forth or side to side motion. The spider was named "Peacock" due to this iridescent, brilliantly coloured and intricately patterned flap.

Occasionally, the male will pause his dancing to tap on the female's head! This complicated routine is what wins her over. Communication goes both ways and if the female is impressed, she will seductively wiggle her abdomen for him and the couple will proceed to get down to business. If not, she will express her displeasure by waving her abdomen back and forth. If the male gets the message, he will put more effort into his dance to woo her. And he must try hard, for there is more than just sex at stake. His very life hangs in the balance! The females, which are much larger than the males, if unimpressed by their efforts, proceed to eat them! This is how sexual selection works. By eliminating the bad dancers with boring flap patterns, only the 'showy' genes are passed on. This gives rise to the handsome, talented spiders we know of today!



Studies have shown that even though vibrations and leg movement play an important role, the visuals and the uniqueness of the flap are what retain the female's interest.

This may be owing to the peacock spider's excellent sense of vision, which they otherwise need to hunt. Unlike other spiders, they don't spin webs to trap their prey and instead, ambush it. And so, for Peacock spiders, a book can indeed be judged by its cover.

Clarita Mendes, S.Y.B.Sc.

NEUROGENESIS IN SONGBIRDS

To a great many poets, the sweet melody of birds in springtime has been a magnificent source of inspiration. It symbolises the vitality that nature has inculcated into all living beings. The significance of this song remained limited to creative literary inspiration until 1984, when scientists Paton and Nottebohm discovered the fascinating phenomenon of adult neurogenesis in the brains of these birds.

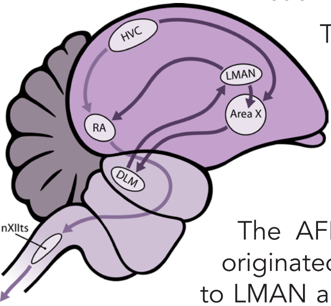
The integration of new neurons in the central nervous system, up until the 1970s, was considered to occur solely during the developmental stages of an organism. When considering the phenomenon of the plasticity of an adult brain, neurogenesis was considered to be improbable and hence kept out the discussion. But this notion changed with the discovery of adult neurogenesis in songbirds.

Songbirds, as they are colloquially known, fall under the sub-order of Passeri (or Oscine) of the order Passeriformes. The most distinguished members being larks (Alaudidae), swallows (Hirundinidae), nightingales (Luscinia), etc. The most notable vocalization of these birds is the melody they produce to woo a mate during mating season (occurring during the spring for most species). Other melodies are produced to indicate to a rival male that a female has already paired or to display territorial ownership. In some species, male and female pairs duet to display the strength of their bond. The common characteristic between songbirds is their large auditory region which allows them to learn and produce their song.

The process of song learning begins with the imitation of the sounds produced by adult birds, by producing a set of incoherent sounds with no communicative intention – a subsong. This subsong is analogous to the noises produced by a human infant during the first few months of its life. In zebra finches, this stage takes place approximately between 28 and 35 days after hatching. As the juvenile birds continue to mimic the sound of the adult mentor, a stereotyped sound that resembles the motifs of the adult song but hasn't reached persistence, begins to emerge. Called as the plastic song, it is much like a child who has learned a language but hasn't mastered grammar. The plastic song eventually leads to the stable song.This song undergoes very few changes over months or years, if at all. In the case of male zebra finch, they attain their song in the first 80 days after hatching.

Zebra finch

There are two distinct pathways that play a part in song acquisition – the posterior descending pathway (PDP) and the anterior forebrain pathway (AFP). Both these pathways originate from a nucleus in the nidopallium (which is a part of the telencephalon) in the avian forebrain called the high vocal centre (HVC). Though both these pathways originate from the HVC, their neurons are different from each other and play different roles when it comes to vocalization



The PDP is comparative to the motor pathway in mammalian brains. It begins from the HVC and extends to the RA (Acropallium). Lesioning of this pathway has shown to affect the production of song in both juvenile and adult males. This pathway was therefore shown to control the song production through controlling the vocal organ, or syrinx.

The AFP, or the recursive loop so called because the pathway originated in the HVC and extends to the RA via collateral projection to LMAN and Area X. When this pathway was lesioned in the juvenile male birds was shown to affect the development of song but not its production, indicating that this pathway is involved in the acquisition of song.

The songs of these birds are dimorphic; the HVC in the male canary is three times larger than in the female and in male zebra finch, it is eight times larger. Research shows that acquisition of the song is under hormonal control. Several studies have shown that the female HVC, when exposed to testosterone, increases in size and the females begin to exhibit songs that are similar to those sung by the males. The seasonal variation in those birds that are seasonal singers show an increase in neurons during the mating season and a subsequent decrease during the summer. It was, initially thought that increase in the volume of the HVC was due to an increase in dendritic synapses. However, it was later shown that there is an increase in the number of neurons as well. The addition of new neurons are constantly replacing old ones. This replacement is seen to be especially present during seasonal changes. It has been observed that male canaries develop a change in song collection during these seasonal changes. Thus, neurogenesis is active in the nucleus HVC and in many areas of the avian telencephalon. This has been shown when lesions were made to the Area X of the adult zebra finch post cell death there was a regeneration of neurons that correlated with the relearning of the stereotyped song post-lesion.

Though much has been discovered since the 1970s, there are still many questions left to be answered. Why replace a whole neuron if effective plasticity can be achieved through a change in the number of synapses and dendritic configuration? Why are neurons replaced frequently in the HVC to RA pathway? Through the study of song learning in birds, it may reveal to us much about memory acquisition, brain repair and, learning in other vertebrates like ourselves.

Salama Udaipurwala, S.Y.B.Sc.

Skylark

IN CONVERSATION WITH
SHASHANK DALVI

Shashank Dalvi completed his M.Sc. in Wildlife Biology and Conservation at WCS-NCBS. He was a core member of the Amur Falcon Conservation Project and the team involved in the discovery of the Himalayan Forest Thrush. He also participated in the Eaglenest Biodiversity Project. He is an a self professed “bird nerd” who loves to travel and is one of brains behind the birding app “Vannya”.

You worked on the Amur Falcon Conservation Project and the discovery of the Himalayan Forest Thrush. Can you tell us about the role you played on an individual level in both projects?

Ramki Sreenivasan, Roko, Bano and I heard about the hunting of Amur Falcons in Nagaland but had no idea about the scale of this hunting, and so we went to Wokha and Doyang Reservoir to investigate. That's when we realised how massive the scene was. There were two main things that we did, first we had to quantify the hunting, so Roko and I went the very next day to the actual place where the hunting was happening; that is, the roosting sites of the Amur Falcons. There, the hunters had set up fishing nets on the trees that the Amur Falcon roosted on. So every morning and evening, when the birds came back to or left the trees, that's when they got entangled in the nets. The first thing I did was to come up with an 'on the spot' scientific methodology to count the number of birds being caught, which we found to be around 18.3 birds per net. Then we counted the number of nets being put up by every hunting group which was around 10-12 nets each from around 60-70 hunting groups. So on an average,12000 to 14000 birds were getting killed each day in the Doyang Reservoir. The second thing I did was the document this hunting, I recorded the entire process and it was these videos which were picked up by Shekar Dattatri, Ramki, Conservation India etc that ultimately triggered the conservation movement for Amur Falcons in 2012 and stopped the hunting by 2013. This project would have been impossible without Bano Haralu, who, being from Nagaland, played a key role in stopping the hunting, along with the Government of Nagaland and the locals. Later, when we started the eco-clubs under this project, I helped design a methodology for social interviews of the children and locals working with us on the project. In two years we saw not only a decrease in hunting but also a change in the mentality of the people living there. This was a very rare success story from the North East and not a single Amur Falcon has been killed in Doyang Reservoir since 2013.


Amur Falcon

The Himalayan Forest Thrush was a completely different story altogether. Dr. Per Alström and I were recording bird sounds in Western Arunachal Pradesh in 2009, May and June, with the aim to record as many bird calls as possible for a particular species. The Plain-backed Thrush is a pretty common bird in North East India and along the Himalayan range, but while recording it's bird song we realized that there was a stark difference in the song above and below the treeline. Below the treeline, the song was was much more musical and very nice, while above treeline, the same bird's song was very scratchy and harsh. Now when it comes to birds, the song evolves much faster than the genes, and so I felt like this difference in song could be the first indicator that there could be a new species hiding in plain sight. So we went and took a look at the DNA from museum specimens and collected more DNA from birds in China, and that's when we realized that there were actually four species of Plain-backed Thrush. The population of birds in China were already classified as sub-species whereas the bird population in India was never officially named as a subspecies of the Plain-backed Thrush and so we realized that India has two species, one being a new species, which became the Himalayan Forest Thrush, named after Dr. Salim Ali. So a lot of work happened simultaneously in the museums, labs et cetera. but my main contribution was it's discovery on the field.

"Your discovery of the Himalayan Forest Thrush was primarily based on its call. How important is bird call as an identification point and what other things do you look out for while trying to identify a bird? Also what technology do you use to record bird calls?"

Birds songs are very important for identification because you always hear a bird before you see it. There may be a lot of instances where you might not be able to see the bird because it's against the light or it doesn't show all the features at all times or something else, but bird songs will always be constant and are one of the most important things to look out for while birding. The first thing I do is listen to the call, then look at how the bird fits into its entire environment, where it's present; for the example the canopy or a small bush, it's movement, it's posture, the shape of the bird with regards to its body proportions and once I look at all these details, then I look at the colour of the bird. According to me, the colour of the bird is one of the least important factors to look at while trying to identify it.


Himalayan Forest Thrush

As for recording calls, I use a Telinga microphone along with a recorder. The Telinga microphone is a directional microphone with a parabola. Taking the microphone with the parabola around on field is very cumbersome but the parabola helps improve the quality of the recording by approximately 15-20 times and so it's definitely worth carrying around if possible.

Besides the Amur Falcon Conservation Project and the discovery of the Himalayan Forest Thrush, you also were a part of the Big Year.Could you tell us about what gave rise to your passion and what led you to commit to something as intensive as the Big Year? Could you also tell us how you believe something like Big Year could help with conservation?

The simplest way to put it, is do things you're passionate about and everything else that follows is easy.

There are a couple of reasons why I did a Big Year. Firstly, I love travelling and observing wildlife and a Big Year was the way to be able to get an idea of what's happening to wildlife across India at a given time. While travelling, you get to know the challenges for different species on an individual, group and a landscape level and this is the first step towards conservation. In the long run I would love to do conservation of birds and other fauna outside national parks as well, and this kind of an experience helps you figure out what is happening across the country with regards to particular species at a given time. This was probably one of the biggest motivations for me to do a Big Year. Also, another important factor was that it was fun! I got to travel a lot, I met a lot of my friends, made many new friends, a lot of people housed me, drove me around. It was an amazing experience. You also meet a lot of like-minded people who are willing to go out of their way to help you out and the only thing you might have in common is your passion for birds. So the experience helps you connect with a lot of people and form a network of people across the country. For example, if tomorrow, you go to the North East to do conservation, taking the help of local birders will help you out a lot. Another thing is that my Big Year was not about just going to a place, seeing birds and ticking them of a list. The most challenging thing about it was being in the right place at the right time. I have never taken a bird guide along with me to the field in my life and I don't like it when people show me birds. I like to go and find my own birds. I decided not to take a single flight to go see any bird and so, almost a year beforehand, I had to sit and plan out the best possible route, so that I could see as many birds as possible in a particular area. And so, my Big Year involved a lot of planning which was fun but also challenging.

You also meet a lot of like minded people who are willing to go out of their way to help you out and the only thing you'll might have in common is your passion for birds, and so it helps you connect with a lot of people and form a network of people across the country.

“So you’ve been pretty much all over India, which place would you say is your favourite for birding? Also having travelled so much you must have seen some pretty spectacular things, which would you say is the most awe-inspiring and memorable moments? And is there anything else left on your life list that you’d like to see?”

Picking my favourite place is probably the most difficult question, but I would have to say the North East and the Andaman and Nicobar Islands.
There are so many amazing experiences to pick from, but one of the recent ones would be when I went to Indonesia. There I went to West Papua and got to see Birds of Paradise. It was truly a spectacle and is one of those things that all birders should have on their bucket list.
There are a lot of birds I’d like to see. One of my boogie birds would be the Orange Bullfinch found in Srinagar. The first time I went to look for it, it was January and because of the cold weather, it had moved to lower elevation levels. The second time I went, it was August and I didn’t have enough time to find it. So I always miss this bird and would love to see it. Another bird I want to see is the Narcondam Hornbill but its really difficult to get to the Narcondam Islands. However, one day I hope to see that as well.

“What would you define as ‘successful’ birding? And how much of this ‘successful’ birding would you attribute to luck, skill and experience?”

The success of birding lies in how you understand the bird. So before you actually go birding, you have to visualise, plan it and know what to expect. 2004 onwards, when I started birding, I went to North East India and there were a lot of birds which hadn’t been seen in the last 70-80 years. For example, the Yellow-throated Laughing Thrush hadn’t been seen in India since 1929 and so Ramki and I went specifically to look for it. We spent around two years looking for this really cool species that no one knew where to find. And this is what I call as successful birding; being able to find resident birds that no one has seen in many, many years, making it much more challenging than finding a new species in an area. So to find these birds, you need to read up on their habitat and then spend a lot of time on Google Earth looking for the right habitat before you actually go out and look for them on the field.
Also, the harder you work, the luckier you get. So the more you look for birds, the more you see. The main part is how you prepare for it. For example, if I’m going for a trip, I will read up on the area, the birds I can see there, read up on them and listen to their calls beforehand. Then you can look at luck, to which I wouldn’t attribute more significance than 5%.

"While working on the discovery of the Himalayan Forest Thrush, what would you say was your biggest obstacle?"

This project happened on a very large scale, all the way from India to China. One of the biggest things in this project was time. We discovered the bird in 2009, but by the time we discovered the species, it was January of 2016. So we had to spend a lot of time in labs, museums and in the field recording the bird songs. Also, we had scattered information about different different aspects of the bird, like information on it's DNA or information on its vocalisation. We had to figure out a way to interrelate and coalesce all these bits of data from the museums, labs and the field and then finally, from all of this ,figure out what the actual identification features are. So the biggest challenge was figuring out which part fits where and then trying to figure out the answer, almost like solving a puzzle!

"Conservation often involves more interaction with humans than with the actual animals. Would you agree?"

For conservation, you have to understand the wildlife, wildlife biology and other academic concepts, but without understanding people, conservation would be very difficult. So yes, I agree with the statement. Sometimes this might seem very boring, tedious or frustrating but that's how conservation works.

Getting back to the Himalayan Forest Thrush, there is supposedly a fourth species, can you tell us something about it?

It has been tentatively named as the Yunnan Thrush after the Yunnan Forest in China. From DNA analyses, we know that there is a fourth species but no one has found the bird, leaving us clueless about its vocalisation and what it even looks like. And, it's not only about looking at museum specimens. The Alpine Thrush and Himalayan Forest Thrush, for example, have very, very minute differences, which you can only look at and observe in the field. So we are still working on it but no one has found the bird in the wild and it will take some time before this happens and the work can proceed.

Could you please tell us something about Vannya and your idea behind it?

I came up with this idea while doing my Big Year. So the idea was to eventually help get rid of heavy books, because a field guide can only give you limited information or limited descriptions as compared to an app which can maybe show you, for example, photographs of the different individuals of a species, different angles of the bird in different light conditions et cetera. It's still a growing app with free access to all. A new version of the app will be released soon with a lot more features such as bird sounds, distribution, elevation and with about another 100 species added to the 1000-1500 species already present on the app.

JACKALS IN A GOAN BACKYARD

This is not a story about a large, charismatic predator in a national park or a sanctuary. This is the story about my neighbours, whom I know so little of. Whom I hear, but rarely see.

Jackals in Goa are often mistaken for foxes and sometimes wolves. In Konkani, 'kolo'- the term for jackals- is wrongly translated into English as fox. The golden jackal or Asiatic jackal forms a part of the semi-urban landscape of many villages in Goa. They persist in human-modified habitats like crop fields, as well as fragmented patches of scrubland, mangroves and hills. Even though jackals can adapt to changes in the land, these animals require natural sites free of human interference for denning purposes. Once in a while, one might come across a jackal crossing a road, a pair of jackals camouflaged against the long stalks of dry grass or even a pack foraging on the garbage dumped in the fields. But jackals are mostly elusive. Their pups and the dens are kept secret. Nightfall is a very important time for animals as human activity is at its least. Jackals are known for howling in packs at night. Two and half years ago, I quit city life to create a new home in a little village in South Goa called Bondorim. We were told by our neighbours that there are "foxes" in the village, although relatively harmless. I did not think much of these "foxes", until I heard the howling for the first time. I pulled my blanket over my head as the strange, long-drawn calls faded into the night. I did not realize then, that the piercing cry that broke my sleep was one of nature's wonders: an indication that there are other living beings living right amongst us in a world that is so often pushed to the edge because of ever-expanding human habitats. What we call urban wildlife, continues to survive even in this hostile, human-dense world.

One reason why jackals howl is to advertise and defend their territories. Howling has a strong seasonal component. During denning season, howling decreases so as not to give away the location of the den to possible predators.

Since June 2017, I have been making simple observations by recording the time I hear packs howl, from the confines of my home. Until the month of August, I could hear the jackals on a daily basis. The howling would begin sometime between 18:30 to 19:45, lasting for a few seconds. Then the pack would howl once or twice between 20:30 and 22:00. During the months of August, the howling reduced. In September, it dropped to almost zero. It is possible that August and September constituted the breeding season. The beginning of October marked the comeback of the loud, piercing cries as the night took over. Howling also functions to attract members of the same pack. The primary social unit of the golden jackal is the mating pair. Both parents take care of their puppies, by digging dens or burrows covered by thick shrubs. When the pups are older, both the father and mother feed their pups by regurgitating partially digested food after hunts. Even after the young are sexually mature, they continue to stay with their parents. So most of the time golden jackals live in small family groups, sometimes just as pairs, but often with the young adult offspring who are helping with the next litter as well.

"When my friends and I ask the locals about them, the common reply is an exaggerated " *Tim sodanch kulio ghaltai...* (They are always howling)" Many times, they would even imitate the calls in an animated manner, followed by embarrassed giggles."

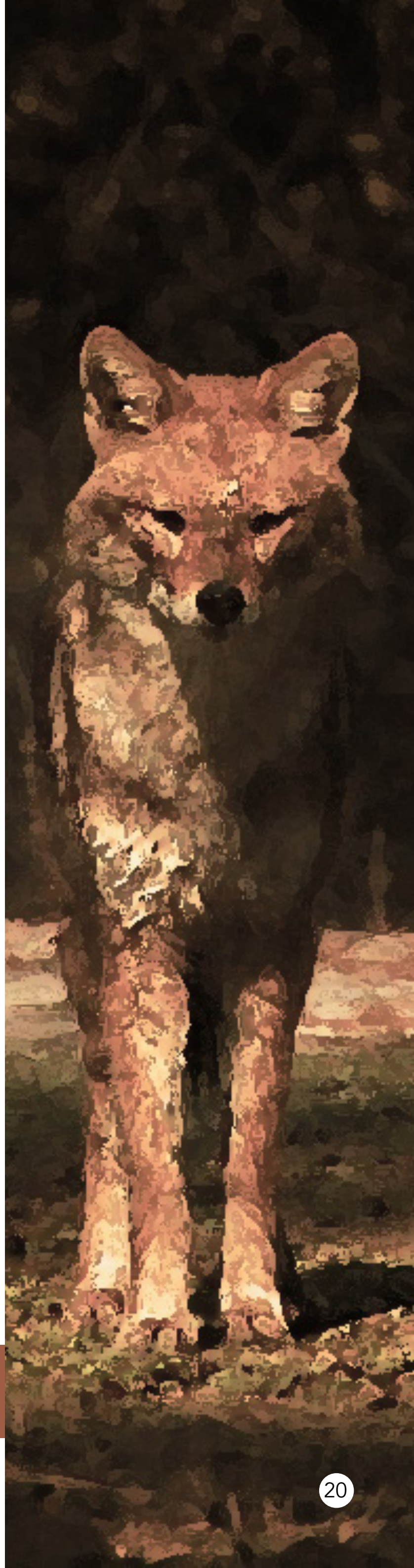
When my college friends and I ask locals about jackals, the common reply is an exaggerated "*Tim sodanch kulio ghaltai*" which translates to 'They are always howling.' Many times, they would even imitate the calls in an animated manner, followed by embarrassed giggles. A migrant worker who works in a dairy farm in Deussua describes the calls in a similar fashion: "Abhi raat mein bahut awaaz karenge aur sham ke chaar baje, kutton ke jaise idhar udhar ghoomenge". - "They make a lot of noise at night and at around four in the afternoon, move here and there like dogs". When humans and wildlife share the same space, there are bound to be interactions of various kinds. The Konkani term for whooping cough is "kolo khonkli", literally translating to 'jackal cough'. The 'whooping' sound associated with the illness has been compared to the cries of the jackals. In some villages, elders previously believed that when a jackal cried, the howls, if heard by small children, could cause them to suffer from kolo khonkli. As a preventive measure, people would blow into the ears of these children as soon as the jackals began to howl. During the interviews, we have also come across people talking about hyenas being present in the village at one point of time. They recall that the calls of the jackals were distinct from that of the hyenas, which was associated with misfortune and death. "*Ganvant kitem vait zata, konui morta, tednam bali rodta…*" - "The hyena would cry if there was a death in the village". In contrast, jackal howls were considered to be a good omen by some.

A mating pair of jackals in a paddy field in Deussua, Goa PC: Malaika Chawla

During field visits to Deussua, my friends and I would wait for the church bell to ring at 18:30. It was especially interesting to hear packs of jackals, from multiple areas, howling together when the bell was rung. This would be followed by domestic and feral dogs howling in response. I consider myself fortunate to be able to listen to jackals not very far from my home.

Despite the fact that the golden jackal is considered a 'least concern' species by the IUCN, its population has undergone significant decline in the recent past in many parts of its geographical range, including India. Golden jackals used to be common throughout the state of Goa. In present times, urbanization of the wilderness and rural landscapes has lead to the decline of urban wildlife. I sincerely hope that we will wake up in time to realize that these sentient beings that we share a planet with, are of utmost importance and must be protected.

Malaika Matthew Chawla, T.Y.B.Sc.
Parvatibai Chowgule College, Goa



DOLPHINS AND THEIR MELONS

The first thing we do when we meet a friend, is greet them. With either a simple word of validation, a handshake or maybe a hug. These behaviours are characteristic of human interaction. But did you ever imagine how the world's 'second smartest' animal interacts?

Dolphins have various ways of communicating on lines that are very similar to humans. Their interactions include a plethora of sounds and bodily contact. They have highly developed organs to modulate and detect sounds in the form of whistles and clicks that are generally at very high frequencies.

Dolphins are capable of extensive vocal communication, which has led to the popular notion that they have a language of their own. This is possible because of an astounding organ that modulates sound production; the melon.

The melon is composed of adipose tissue which acts as a sound lens to focus the dolphin's vocalizations. It is present between the blow-hole and the tip of the snout. Due to its believed bioacoustic properties, it is hypothesised to also help in the organism's echolocation. The fact that the melon is of irregular composition and is surrounded by air sacs, suggests that the sound waves are emitted in all directions with varying intensities.

This is a very helpful trait for dolphins because they hunt in groups. Hunting packs of orcas and bottlenose dolphins have excellent interactive maneuvers amongst individuals. They "talk" by clicking, locate prey using echolocation and sometimes even use directive frequencies to confuse shoals of fishes. This isn't surprising, given that hunting packs hardly ever fail at obtaining food. The hunt of the dolphin is one of the most well planned and executed group activities seen in nature.

The melon is a wonderful example of evolutionary brilliance and the reason dolphins are capable of having their own high pitched language. It is wonderful to know that one of the most loved aquatic animals are in fact one of the few organisms to possess a language and an organ that has evolved in accordance with the need of communication. The dolphin with its melon, is a classic example of how intricately the denizens of animal kingdom communicate.

Like humans have names, dolphins have whistles unique to individuals.

Around the beginning of the 21st century, marine biologists obtained a sound clip of a high pitched sound wave much similar to those made by toothed cetaceans. It was so perfect, that they thought it belonged to a species that had highly evolved bioacoustic properties. The organism, if existed would've had a much more complex melon than the ones found today. They called it "The Bloop"

Around the same time, it was theorized that mermaids existed. This theory gained massive backing due to the existence of The Bloop. It was later proved that the sound probably belonged to an underwater volcanic eruption, so the mermaid theory was thrown out the window.

However, it is interesting to note that this gave a massive push to cryptozoologists on their take on communication among mermaids. It is a thought to ponder upon over a nice cup of coffee.

Kartikeyan Premrajka, S.Y.B.Sc.

CRETACEOUS ROAR

How intrigued were you on watching your first Jurassic Park Movie? Or taking a trip to the local museum and encountering a humongous prehistoric fossil? Or just hearing the term 'Dinosaur' for the first time? Imagine a 40 feet monster in your backyard, roaring at the top of its voice and shattering not only the decibel limit but also your senses! This is a scenario which would have been true, if you could travel 65 million years back in time to when these creatures ruled the planet. Dinosaurs fascinate us all as they were mysterious, intimidating and monstrous on appearance. Just like us humans, they too communicated with one another using a myriad of tools.

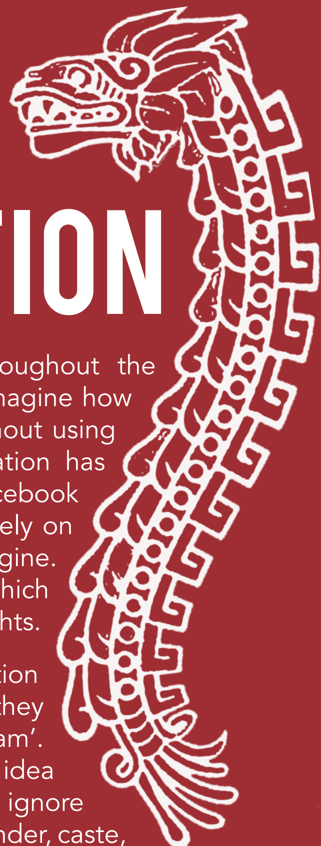
Birds are considered to be living relatives of the dinosaurs. Modern birds descended from a group of two-legged dinosaurs known as theropods, whose members include the towering Tyrannosaurus rex and the smaller velociraptors. Birds produce sound through a specialised vocal organ called a Syrinx which functions as a 'voice box' and is located at the base of a bird's trachea. This unique bony structure is what allows crows to caw, robins to sing, and parrots to talk like people do. The fact that this structure is unique to birds suggests it evolved after early avians split from other dinosaurs. We can safely assume that a T-Rex didn't chirp!

So how do we find out what these fearsome creatures may have sounded like? Alligators and crocodiles may come to the rescue here. Even though they are the more distant relatives of the dinosaurs, they are still the closest living cousins to the dinosaur group as a whole. They create a range of deep-throated sounds through the larynx, which is a more archaic vocal setup and even though it doesn't have the same punch as Steven Spielberg's famous roar, it is probably a lot closer to what the real T-Rex sounded like. Recent studies suggest that the T-rex most likely had a larynx just like the present day alligators or crocodiles causing them to make booming, hissing or rattling noises while communicating with each other. Scientists call these sounds as "Closed-mouth vocalisations" as they are low, full-throated noises. This sound is produced in the present-day alligators by passing air through constricted openings such as the nostrils, glottis and palatal valve.

Along with a resounding call, the T-Rex also possessed an incredibly strong sense of smell due to its large olfactory lobes in the brain that would have allowed it to sniff out prey from hundreds of feet away even in the dark, identify other potential competitors or navigate back to its nest. These along with its dominating figure, razor sharp teeth and a top speed of 19 km/hr (for an organism weighing 8000 Kgs!) made the T-Rex one of the greatest predators to have ever walked the planet.

Ritabrata Ghosh, F.Y.B.Sc.

MILESTONES IN COMMUNICATION



Human communication has evolved tremendously throughout the millions of years that humans have existed. It's hard to imagine how our ancestors used to communicate with each other without using words, pictures, and heck, memes. Human communication has evolved from pictograms to words to tagging friends in Facebook posts, which, as humorous as it may sound, is true. We rely on more means to connect with each other than we can imagine. I'm going to walk you through some major milestones, which over millennia, have shaped how we convey our thoughts.

Pictograms are the earliest confirmed form of communication amongst the human species. Now, before I tell you how early they originated, let me clarify the meaning of the word 'pictogram'. A pictogram is essentially a graphical representation of an idea that can be universally understood. Here, we completely ignore the concept of context, which means that no matter the gender, caste, religion, nationality, and such attributes of the person, they should find the same meaning in a pictogram. A pictogram is not the same as a 'sign' or a 'symbol', since signs can be exclusive to a group of people. For instance, when we Indians join our hands to say namaste, it makes sense to people of an Indian background, but it may not always signify something to a Canadian. 'Namaste' is an example of a sign. Back to pictograms; the earliest signs of them being used date back to 30000 BC, where caves were painted with symbols that represent people and animals. Still, pictograms didn't become popular until more 'modern' times, that is, 9000 BC. Pictograms in ancient times were of two kinds- petroglyphs and petrographs. Petroglyphs are symbols carved into surfaces like walls or floors, whereas petrographs are symbols painted on them. Now, you may ask, why are pictograms so important? Well, think of logos. They are essentially pictograms! Say you walk into a McDonald's outlet. How do you recognise the employees? You look for someone with the classic yellow 'M' on his or her t-shirt! How do you locate Whatsapp on your phone? You look for a green logo shaped like a comment-box with a white imprint of a phone at its centre! We use pictograms so often and don't even realise it.

Alphabets are the basic building blocks of most languages known to man, and so they have been for many millennia. By around 17-1800 BC, people had started using sounds to communicate. It was like a primitive language, except that it couldn't convey a whole lot. By that time, people had also started to realise that petroglyphs and petrographs could communicate very few, and very vague ideas. A means to achieve detailed communication was required, and so a group of Semitic-speaking people (people of an Afro-asiatic origin) created 22 symbols, inspired by Egyptian hieroglyphics. These symbols are probably the first recorded consonants. An obvious question comes to mind…what about vowels? Well, since language hadn't developed so much at the time, people could figure out meanings of clubbed consonants. It wasn't that hard to understand, since the meanings of those 'words' got rooted into people's minds as they started using them more regularly. Years after this alphabet was introduced, it reached Greece, since merchants had started using this Semitic alphabet to communicate with each other. The Greeks adopted this alphabet, made their own versions of it, and also introduced vowels. They believed that the sounds between the consonants were very important, since they would reduce ambiguity as the language developed. Well, it seems like we've gone back to those times now. All of us text things like 'brb', 'np', 'idk', and so on, but understanding the meanings of words really is the foundation of understanding any language. Numbers have very similar origins, and we've all had history lessons about those at some point in our lives, so let's move on.

"They believed that the sounds between the consonants were very important, since they would reduce ambiguity as the language developed. Well, it seems like we've gone back to those times now. All of us text things like 'brb', 'np', 'idk', and so on..."

Now, I'm going to skip a few millennia and focus on something we wouldn't be able to live our lives without nowadays: the **telephone.** There were many milestones in the time period, like pigeon post, the typewriter, and the telegraph, but I want to focus on milestones that probably matter a little more to our generation. Well, back to the topic at hand. Alexander Graham Bell is recognised as the inventor of the traditional wired telephones that are still used today. The history of the telephone is quite interesting. Its design was inspired by the telegraph, which could send a single message along a wire over large distances. Bell believed that he could improve the telegraph, by introducing a mechanism that allows him to send multiple signals of varying frequencies over a wire simultaneously. Long story short, Bell succeeded in doing so, and got the patent for it in the year 1876. Within the next decade, he also succeeded in transmitting messages using a beam of light as a medium. This would later evolve into what we now call 'fibre optics'.

Quick question: Which company do you think developed the world's first mobile phone? The answer is probably not what you guessed. The only

challenge in making a telephone mobile was portability, and Cooper achieved it. It may not have been the finest mobile phone in history, since it weighed a little over a kilo the company Motorola invented the world's first mobile phone. Specifically, a senior engineer called Martin Cooper did. The only challenge in making a telephone mobile was portability, and Cooper achieved it. It may not have been the finest mobile phone in history, since it weighed a little over a kilo, took 10 hours to charge completely, provided half an hour of talk-time, and was as big as today's average telephone receiver, but it changed the world. Motorola went commercial with an improved model of Cooper's phone, and called it 'Motorola DynaTAC'. At that time, its cost was $3995 (around 260000), and was made only for the elite members of society. In 1997, Nokia released the 'Nokia 6110', and this is where the classic game 'Snake' was introduced. In the year 2000, they released the 'Nokia 3310', which is the model people call 'indestructible'. By this time, mobiles had reached the common man. Five years later, the 'Blackberry 7270' hit stores, and was one of the first phones to have Wi-Fi functionality. Why am I telling you all this? What do we have to gain from learning the origins of some old-fashioned mobile phones? Or words? Or pictograms?

Well, these inventions quite literally changed the way the world worked. So many inventions could take place because of advances in human communication. Name anything that is manufactured now, and it wouldn't exist if it weren't for pictograms, words or numbers. Humans are probably the weakest animals in the animal kingdom. We consider ourselves powerful, smart, dangerous, which we undoubtedly are, but only because we have all these modern resources at our disposal today. We probably would be extinct if a less-evolved ancestor of ours hadn't decided to carve a symbol of an animal on a wall, or if a merchant hadn't decided that sounds could be given meaning and crafted into words. Our history is rich, and full of unexplored wonders, and it doesn't hurt to be grateful for what was passed down to us, or to be inspired to take it forward.

Pratyush Panshikar, T.Y.B.Sc. I.T.

SOUND SCIENCE AND BAFFLING BEHAVIOUR

Why make sounds?

Finding a mate can be difficult, especially for animals that are small, short-lived and there's only a few of. Many insects answer to this description, and can face a severe problem as a result. If the population density drops to a level where individuals can barely find a mate, many might not be able to reproduce. This reduced biological fitness, which is a result of low population densities, rather than adaptive value, is sometimes called the Allee effect.

Many insects face up to this existential crisis with some élan; they produce impressive signals which travel much further than they can, making them much more conspicuous. Insects produce sounds (crickets), light (fireflies) and even smells (moths) to indicate their presence and willingness to mate. An added benefit to broadcasting signals is that they can transmit information about the position, species identity and even quality of the signaller. Thus, reproductive isolation and sexual selection can begin long before potential mates have even made contact.

Sound is one of the most interesting signalling modes. It's lower in energy than light and thus easier to produce. Sound can also travel around obstacles, unlike light which is easily blocked, and, unlike pheromones, it carries information about the location of the signaller. Many a group of insects have decided on using sound as their primary mode of communication, but the most prominent to human ears are crickets. Crickets belong to the order Orthoptera, which consists of two subfamilies, the Caelifera, (grasshoppers, who also sing but usually in the day) and the Ensifera or crickets who sing mainly at night in the tropics.

If cricket sounds are predominantly a way to attract mates, then these signallers must be attempting to make the most conspicuous possible signal. How do they do this? How do these tiny insects make themselves so loud? There are many species of crickets and most of them share one common mechanism to achieve this goal: mechanical resonance. The body parts they use to sing, their wings, 'ring' at a very specific frequency, much like a tuning fork does, making them louder than they would otherwise be. There is one group of crickets that goes a step further though: tree crickets build a tool to make themselves louder!

Why make baffles?

To make sounds, tree-crickets like all other crickets, rub their wings together. This sets them into resonant vibration. As the wings vibrate, their back and forth motion creates small local changes in air pressure i.e. creates sound. For example, while vibrating, as the wings move in the forward direction, they compress the air in front of them and rarefy the air behind them, and vice versa

There is a problem, however. The compressed and rarefied air meets at the wing edges, mixes and the air pressure equalises. This effectively cancels the sound out. The effect is often whimsically called acoustic short-circuiting. The smaller the wings, the larger the sound cancellation effect and the less efficient the cricket's efforts. And tree cricket wings are barely a centimetre long. So how do they get so loud?

To work around acoustic short-circuiting and to make themselves louder, some tree crickets make a tool called a baffle. They neatly cut a hole near the centre of a leaf and then sing from inside this hole, with their wings flat against the leaf surface as seen in thei mage accompanying the title. Wing vibration occurs like before, but the baffle separates the front and back face of the wings. This forces the sound radiating from the front of the wings to travel all the way to the leaf edge before meeting the sound radiating from the back, reducing the short-circuiting it experiences, making the cricket louder.

"Despite their tiny brains, the crickets appeared to already know and follow all the rules we had painstakingly discovered: they always picked the largest leaf, made a perfectly sized hole and placed it close to the centre. Tree crickets were capable of making the best possible baffle!"

Not all baffles work equally well and recently, we decided to find out if male tree-crickets, despite their tiny brains knew how to make the best possible baffles. To study this problem, first we had to figure out what the best possible baffle was. We measured the wing vibrations and sounds of real tree crickets, and used them to make a simulation model of a cricket singing from different baffles. In our model, we could change many different things about the baffle itself, while keeping the singing effort of our model cricket constant. Something that we could never have done with a real live cricket. From our models, three simple rules emerged that would lead to the best possible baffle: use the largest available leaf, make a hole the size of the wings, and place it at the centre of the leaf.

Next, we put the crickets to the test. We did a series of experiments: we looked at their singing and baffling behaviour in the field, and then did some very controlled behavioural experiments in the lab. The most revealing result was from an experiment in which we offered the crickets a choice between two differently sized leaves. Amazingly, we found that in every single case, the crickets chose the larger leaf! Despite their tiny brains, the crickets appeared to already know and follow all the rules we had painstakingly discovered: they always picked the largest leaf, made a perfectly sized hole and placed it close to the centre. Tree crickets could make the best possible baffle!

> "What was even more surprising was that the crickets made a near optimal baffle in a single attempt, they didn't require any trial and error to do this. Insects are generally seen as only being capable of gradually improving the structures they live in and never quite reaching optimal efficiency."

What was even more surprising was that the crickets made a near optimal baffle in a single attempt, they didn't require any trial and error to do this. Insects are generally seen as only being capable of gradually improving the structures they live in (nests, webs, or burrows), never quite reaching optimal efficiency. So this finding was something of a surprise. When we thought about this problem a bit more through, we realised that not all objects can be changed progressively. For instance, you can shorten a stick by breaking pieces off it, but you can't lengthen it the same way. This is certainly the case with baffles, a hole in a leaf cannot be erased or moved; only a new one can be made. A new hole is more work, of course, but worse, the leaf is irreversibly changed and we found that this new holed can even make the baffle less efficient. So as far as baffles are concerned, making them in one shot is the best option. No wonder, these tiny brained crickets have evolved a behavioural routine that lets them get it right the first time!

This work highlights something interesting not just about the cognitive aspects of communication, but also about insect cognition. Insects are small, and fairly easy to underestimate. It is easy to see apes and crows as clever animals that learn to use tools and solve difficult problems with their large brains. However, it's very hard to imagine that any insect could ever approach a similar degree of intelligence. Yet here is an insect that can optimize its use of a very tool-like object! Intelligence, it turns out, is a slippery concept, and there is nothing fundamentally different about insect brains other than their size. Indeed, it may be best to invert the problem and ask how insects do all that they do despite their smaller brains and lower computational power.

MALE AND FEMALE TREE CRICKETS

Indeed, insects have done a lot! They been crawling around the earth for more than 400 million years. They have survived in an evolving and changing world: shifting diets as plants crossed over to land and went from making spores to growing flowers and fruits. They have thrived even as the continents drifted, and the atmosphere and climate changed and have lived through various mass extinctions. They've achieved an immense diversity of habits and invaded nearly every habitat on earth, equipped with only their minute brains to make sense of the world they encounter. Hidden away in their tiny nervous systems are innovations in computational problem simplification that we would do well to discover as we start to build our own minimal and autonomous robots.

Natasha Mhatre studies acoustic communication in insects. She is a postdoctoral research fellow at the University of Toronto, Canada under the department of Biological Sciences, Scarborough. She is an expert in Animal Behaviour, Animal Physiology and Acoustics.

PHOTO OPS

Costa's Hummingbird - Amartya Tashi Mitra

Deccan Banded Gecko - Sanchit Mishra

Blackbuck and Laggar Falcon - Harshit Singh

Vine Snake feeding on a Raorchestes frog - Arjun Kamdar

Huntsman Spider with prey - Michael Doe

Hummingbird Hawk Moth - Sunal Kumar Roamin

Wild ponies in Long Mynd, Shrophire, England - Eleanor Ryder

Tal Chappar, Rajasthan - Puneet Verma

THE SECRET LANGUAGE OF MANTIS SHRIMP

Mantis shrimp are marine crustaceans that belong to the order, Stomatopoda. They are among the most important predators in many shallow, tropical, and sub-tropical marine habitats. In terms of communicating privacy, mantis shrimp are way ahead of humans. Their security measures are built in biologically and involve a peculiar communication strategy based on circular polarised light.

Although Mantis shrimps are only around six inches long, they are extremely strong. In fact, these crustaceans have the fastest punch in the animal kingdom. They repeatedly strike using a club-like appendage to break open the shells of their prey. What's more is that the appendage can withstand these blows and has thus inspired the designs of suits of armour and football helmets. Another interesting fact about Mantis shrimp is that they possess one of the most elaborate visual systems and are known to have 12 colour-receptive cones (compared to the three that humans have). They can move each eye independently and can also see infrared, ultraviolet and polarized light.

The Mantis shrimp is under constant observation and researchers have been discovering new facts about their complex nature and unique features. Researchers from the Queensland Brain Institute at The University of Queensland discovered a new form of "secret light communication" used by mantis shrimp.

> "Mantis shrimp punches produce flashes of light on impact. This happens because the speed of the club causes the pressure of the water around it to lower and this causes the water to boil! "

The team found that tiger mantis shrimp (Gonodactylaceus falcatus) can reflect and detect circular polarizing light, which they use "as a means to covertly advertise their presence to aggressive competitors". Professor Justin Marshall stated that they had found out that a Mantis shrimp has the ability to display circular polarised patterns on its body, mainly the legs, head and its heavily armoured tail. He also mentioned that those three regions were most visible when the shrimp curls up during conflict.

To observe this, researchers put one shrimp into a tank which had two burrows it could hide in. One of these burrows reflected unpolarised light and the other one reflected circular polarised light. Observation revealed that the mantis shrimp chose to hide in the burrow which had unpolarised light, around 68% of the time.

“Mantis shrimp could possibly help doctors detect cancer. This is because cancer cells, unlike healthy cells, don’t reflect circular polarised light. Something that mantis shrimp eyes can easily detect”

Professor Marshall also stated that if you label holes with circular polarising light, the shrimp will avoid going near them, since they infer the presence of rival shrimp!

In addition to proving that mantis shrimp are even more fascinating than we had presumed, these findings also have medical applications. According to Professor Marshall, they could possibly help doctors detect cancer! This is because healthy cells reflect circular polarised light, unlike cancer cells. Since humans can’t naturally detect circular polarizing light, special cameras equipped with sensors may detect cancer cells much before the human eye would be able to.

This study provides the first evidence for the use of circular polarised light as a visual communication signal in any animal. To conclude, we now know that the mantis shrimp prefers to occupy burrows emitting light of the same wavelength and intensity without the circular polarised light and that researchers hypothesise that this method is used as a natural response to implicate a signal for burrow occupancy, thus indicating privacy.

Janvi Gandhi, S.Y.B.Sc.

COMMUNICATION IN THE DARK OF THE NIGHT

In daylight, animals show an extreme diversity of communication strategies, visual signalling being one of the most obvious and widely used. Variability in colouration is a particularly common signal, and bird plumage is one of the best examples. At sunset, however, colours become progressively more indistinguishable, and thus useless for signalling. Scientists have long assumed that crepuscular and nocturnal animals forgo such visual signals and rely solely on sound. But animals have shown to be able to use elaborate signalling to communicate with conspecifics and heterospecifics in different contexts and for a variety of reasons.

Twilight and nocturnal light (moonlight and starlight) may have acted as evolutionary forces, shaping different methods of visual communication among a broad range of animal species. The need to convey information to conspecifics by visual communication in nocturnal species may have promoted convergent evolution towards white visual signalling at crepuscule in distantly related groups of nocturnal species. Indeed, unpigmented fur and feathers are ideal for signalling at twilight, when contrast is more important than colour. Indeed, contrary to common wisdom, recent research shows that many crepuscular and nocturnal species use a surprisingly large repertoire of visual signals to communicate in a variety of contexts, even in the dark.

"Indeed, contrary to common wisdom, recent research shows that many crepuscular and nocturnal species use a surprisingly large repertoire of visual signals to communicate in a variety of contexts, even in the dark."

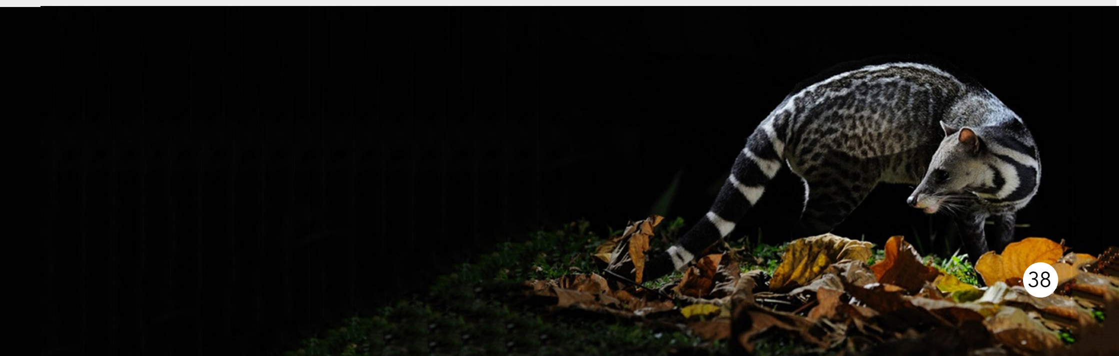
There's every reason to suspect that visual signalling is more widely employed by nocturnal animals, including some birds, mammals, and frogs, than previously thought. A number of nocturnal species bear patches of white fur and feathers and make twilight displays, including e.g., the great horned owl, the great grey owl, the little owl, the great snipe, and various species of bustards and nightjars. Recent research has highlighted several lines of evidence, showing that visual communication has been an overlooked communication channel, although it does not seem to be as generalised as vocal communication. In any case, the general rule seems to be that animal conversations involve visual, as well as vocal, dialogue.

Most crepuscular and nocturnal species communicate via voice, which is the reason why we have thought for decades that most of them rely exclusively on vocal signalling. In addition, many mammals have glands that allow for chemical communication, but few of them seem to have evolved visual communication. Such an adaptation in a narrow range of species might indicate that very specific conditions and/or needs are required for visual signals to evolve in nocturnal species. However, the reasons behind the evolution of visual signalling still remain a mystery.

We need to improve our understanding of the way visual signals respond to ambient light and the peculiarity of the vision system of those species that communicate by visual signals. Indeed, understanding visual signalling starts with the knowledge of how animal eyes are adapted to see and interpret such signals. Because the environment plays a major role in the expression of animal communication, it is also crucial to understand the success of this way of communication as a function of environmental conditions. Understanding how animals perceive natural environments under dim light will allow us to gain insights into how they have adapted their communication. The dim nocturnal world seems to be much richer in visual information than we have previously given it credit for and we, human eavesdroppers, must listen and watch.

Vincenzo Penteriani is known as 'Mister Eagle Owl'. He is currently a researcher in the Spanish Council for Scientific Research (C.S.I.C.), Pyrenean Institute of Ecology (IPE) and Department of Biodiversity and Restoration, Spain. He has studied eagle owls, brown bear ecology and behaviour and movement ecology

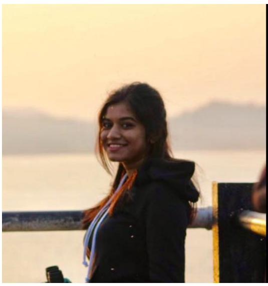
María del Mar Delgado currently works as a postdoctoral researcher in the department of Biological and Environmental Sciences at the University of Helsinki. Her research areas include evolutionary biology, animal dispersal, population dynamics, community dynamics and conservation biology.



INTERNSHIPS

Every year, the students of the Zoology department set out to the field or the laboratory to apply the knowledge they have acquired over the year, to gain some hands-on experiece and to help contribute to the field of research.

In May 2017, 3 students, Dolsy David, Wenzel Pinto and Jason Coutinho interned for a project led by Mr. Gaurav Vashishtha who is pursuing his Ph.D. at the University of Delhi. They worked on population estimation, intra-population interaction and anthropogenic stress on the Gharials of Katarniaghat Wildlife Sanctuary, Uttar Pradesh, funded under 'Rufford Small Grants"

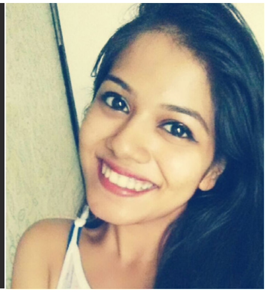


Shafaq Teli worked under Dr. Harish Chafle; a bronchologist and sleep disorder specialist. She spent 3-4 hours at his clinic daily, observing and helping him whenever needed as he treated his patients. She was also given the opportunity to attend a seminar on minor surgeries along with Dr. Chafle. During the course, she learnt a great deal about bronchology and clinical pathology.

Sudipta Kalita did his internship under Mr. Nilkamal Choudhury, on estimating the bird biodiversity in eight wards of the Bongaigaon municipal area, Assam and on raising awareness about said diversity among the local people. The data he helped obtained facilitated the identification of habitat sinks of pigeons and house crows and important nesting sites of birds belonging to the family Ardeidae, which comprises of herons, bitterns, egrets et cetera.

Jacinta Noronha worked at Dr. Bhagwat's Veterinary Clinic, Ratnagiri. She interned for a month, from April to May 2017, during which she gained knowledge about various disorders, animal parasitology and also got the chance to assist him in few minor and major surgeries. She also learnt about different vaccination techniques including intramuscular and subcutaneous types.

Shyla Shyamsundar completed her summer internship under the guidance of Dr. Madhavan K.V., a veterinary surgeon and radiologist at Sri Chamarajendra Zoological Gardens, Mysore. Her work revolved around captive animal management and zoo administration. She gained invaluable knowledge about animal acclimatisation, rehabilitation, diet and health care. She was also witness to animal surgeries, post mortem and vaccination procedures.



Zachary Borthwick interned with Phoenix Veterinary Specialist where he got the opportunity to speak with 'pet parents' inquiring about their pet's behaviour, diet and quirks. The data that he managed to collect from these discussions was used by the veterinarian to link ailments and physiological problems to their peculiar behaviour.

Shimontika Gupta interned at the Agumbe Rainforest Research Station and worked on the breeding ecology of Yellow-wattled Lapwings; documenting and observing their behaviour over different stages of their breeding cycle and performing an in – depth habitat survey of their breeding grounds.

LISTENING TO WHALES

The Earth is full of mysteries and one of the most intriguing of these are whale songs. These large mammals are renowned for producing songs of astounding beauty, complexity and duration. Despite decades of research, scientists are still unsure why these humongous creatures engage in remarkable acoustic displays.

From high wails to deep growls, rhythmic scratches and tearful moans, whale songs encompass a wide range of emotions in some of the longest calls ever performed by an animal. A single tune can last for nearly 24 hours at a stretch and people are often moved to tears by their calm yet intense melody. Whale songs have influenced classical, pop, folk and jazz music, and have been credited with giving rise to a global movement to conserve these astounding creatures. The first recordings of whale songs were made in the 1960's by the US Navy using hydrophones. When they were heard by the rest of the world, people were left in awe. The National Geographic produced the album; "Songs of The Humpback Whale" in 1970, which became one of the bestselling environmental albums of all time.

The clicking sounds produced by dolphins and sperm whales are used to communicate and coordinate foraging activities. Marine biologists have analysed whale songs by breaking them down into components. These acoustic displays are made up of "units" which are defined by marine biologists as the shortest sounds that still seem continuous to the human ear. These units are comprised of wails, moans, and shrieks. Songs have also evolved and changed over time. Sadly, human activity is disturbing marine life via acoustic pollution. Human made sounds interfere with cetacean communication and may even cause beaching. Sources of this pollution include boats, oil rigs, seismic airguns, underwater explosions, construction, pile driving, acoustic deterrent devices, and scientific and military sonar systems. Whales can be influenced by sounds produced as far as 200 km away. It is up to us to save these mystical species and their artistic symphonies.

Mrunal Deolalkar, S.Y.B.Sc.

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Image Courtesy: Flickr, Livescience

COMMUNICATION IN THE DARK

Image Courtesy: 123RF

Special Thanks

This is Imprint's third consecutive year in production and although each year teams change, and we bring out a new theme every time to our readers; the spirit for spreading accurate knowledge and improving aptitude for zoology is a mainstay for us. Something else that has been uniform similarly are the striking cover designs each year. All of this is thanks to Siddharth Bhatia, a BMM alumni from our college and professional designer who has painstakingly made these for us year after year. Here's a shout out for his dedicated efforts and for those interested in his work, you can look up some of Sid's incredible designs on his instagram handle- **siddychan**

We would like to thank Harshit Singh, an alumni of the Zoology Department and Student Editor for Imprint-2017, for his invaluable contribution to the designing and proof-reading this publication.

Department of Zoology
Xavier's Zoology Association